Marwadi University
Marwadi Chandarana Gr

Faculty of Engineering and Technology
Department of CSE – AI, ML and DS

PROJECT REPORT

A I-Assisted Web Application Project

Semester 3 – CSE (AI/ML)

Project Title

[AI Study Planner]

Submitted by

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1. Introduction

Briefly introduce the real-world problem and the application developed. Explain how AI tools and prompt engineering were used during the project. [Write 1–2 paragraphs.]

2. Target User

Identify who will use the application. Examples: students, teachers, patients, citizens, customers, players, etc. [Describe your target users.]

3. Tools Used

Mention the technologies, languages, libraries and development environment used. Mention each AI tool and what it was used for. Examples: idea generation, requirements, UI design, code generation, debugging, testing,

4. Prompt Engineering Process

Include prompts : Write purpose of prompt and put **screenshot of prompt**

1. Idea/problem analysis
2. Requirements/features
3. UI/layout design
4. Code generation
5. Testing 6. Debugging and improvement

5. Application Design and Development

Describe the page structure, navigation, major UI components and application flow.

- Insert screenshots/diagrams below.
- Insert system/page structure diagram
- Insert UI screenshots with captions

6. Final Application

Describe the final working features and the final user experience. Provide the link of application.

7. Conclusion

Summarize the project, the role of AI, the final result and possible future improvements.

1. Introduction

Students often face difficulties managing their daily study schedules, completing assignments on time, and preparing for examinations. They may not know how much time to give to each subject or how to balance college studies with competitive exam preparation. Poor time management can lead to stress, incomplete tasks, and ineffective learning.

To solve this problem, we developed an AI Study Planner, an application designed to help students organize their studies and manage time effectively. The application allows users to enter their subjects, available study hours, priorities, and examination dates to create a personalized study plan. AI tools were used during the project to generate ideas, identify requirements, design the user interface, generate code, and improve the application through testing and debugging.

2. Target User

The target users of the AI Study Planner are students who want to improve their study habits and manage their academic work efficiently.

The main target users include:

- College students: To manage semester subjects, assignments, and examinations.
- School students: To organize daily homework and revision.
- Competitive exam aspirants: To plan preparation for exams such as GATE, JEE, and other entrance examinations.
- Students with busy schedules: To divide their available time between multiple subjects.

The application is designed for students who need a simple, flexible, and easy-to-use study planning system.

3. Tools Used

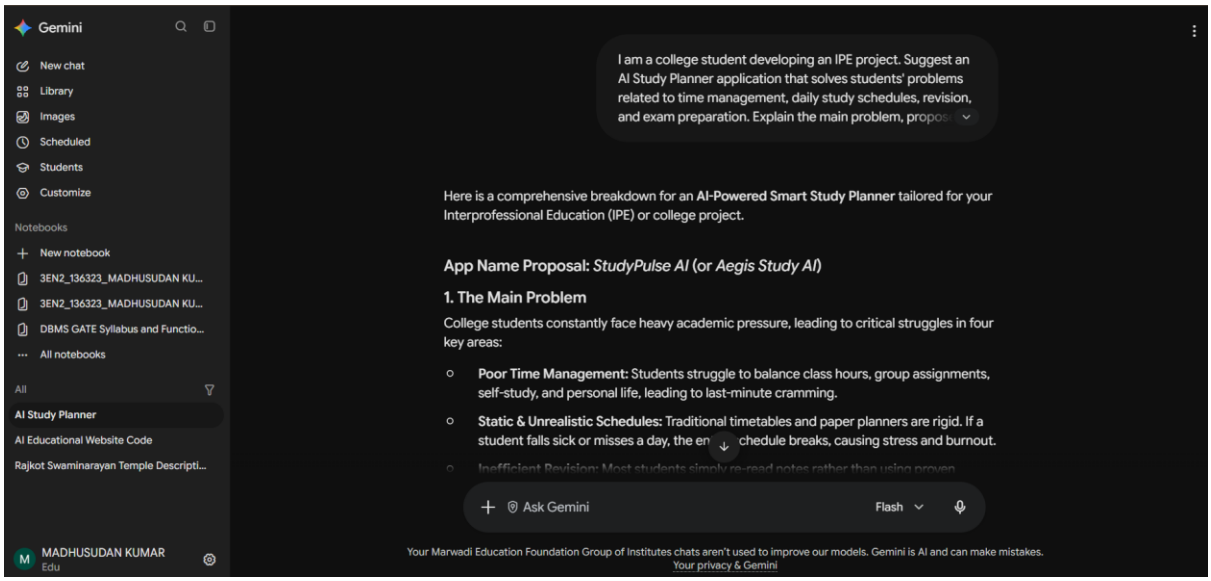
Tool/Technology	Purpose
Google Gemini or ChatGPT	Idea generation, requirement analysis, prompt writing, and problem-solving
Google AI Studio or another AI coding platform	Generating and improving application code, if used
HTML	Creating the structure of the application
CSS	Designing the layout, colors, buttons, and responsive interface
JavaScript	Handling user input, study-plan generation, and interactive features
Web browser	Running and testing the application
Screenshot tool	Capturing the application's interface and testing results

Prompt 1

Purpose: To identify the problem and decide how the application can help students.

Prompt:

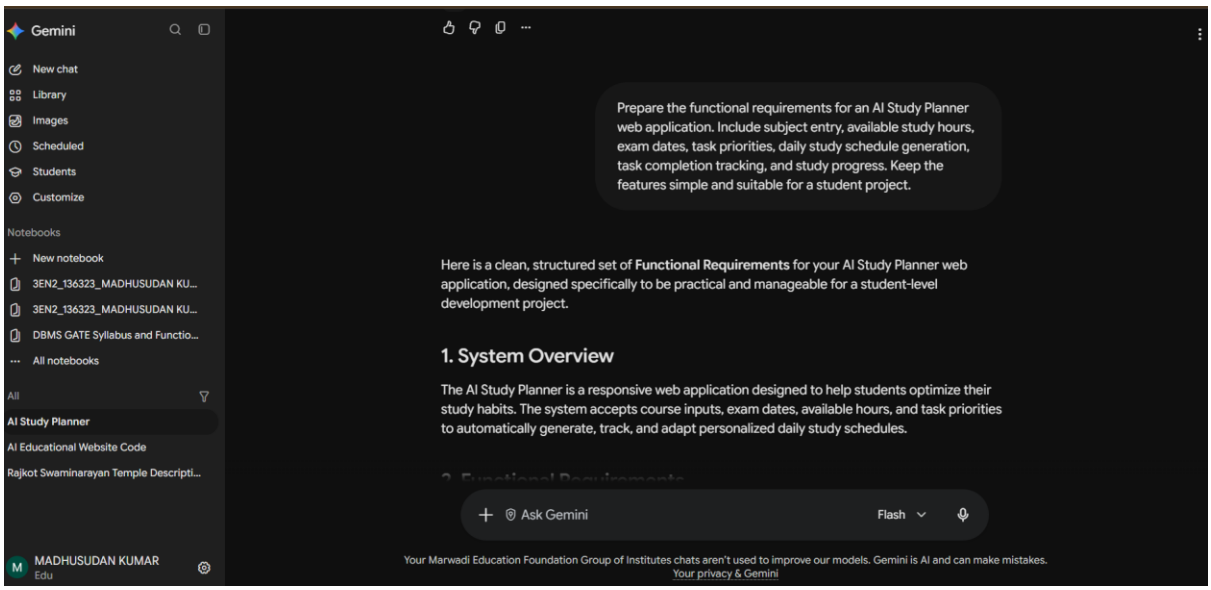
I am a college student developing an IPE project. Suggest an AI Study Planner application that solves students' problems related to time management, daily study schedules, revision, and exam preparation. Explain the main problem, proposed solution, target users, and important features in simple English.



Prompt 2

Purpose: To identify the functional requirements of the application.

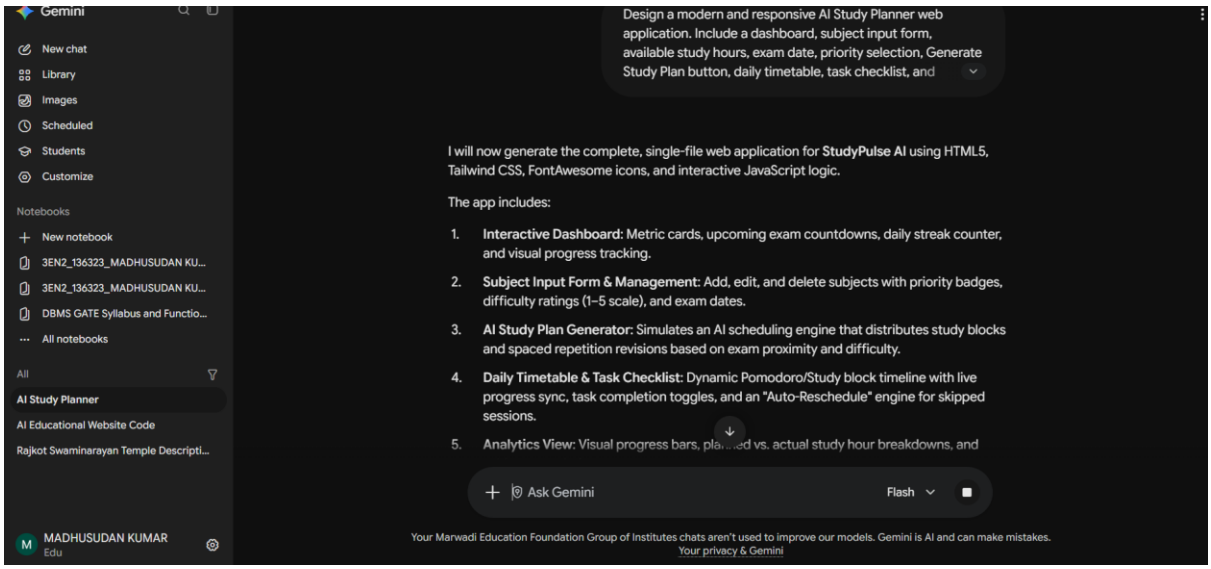
Prompt : Prepare the functional requirements for an AI Study Planner web application. Include subject entry, available study hours, exam dates, task priorities, daily study schedule generation, task completion tracking, and study progress. Keep the features simple and suitable for a student project.



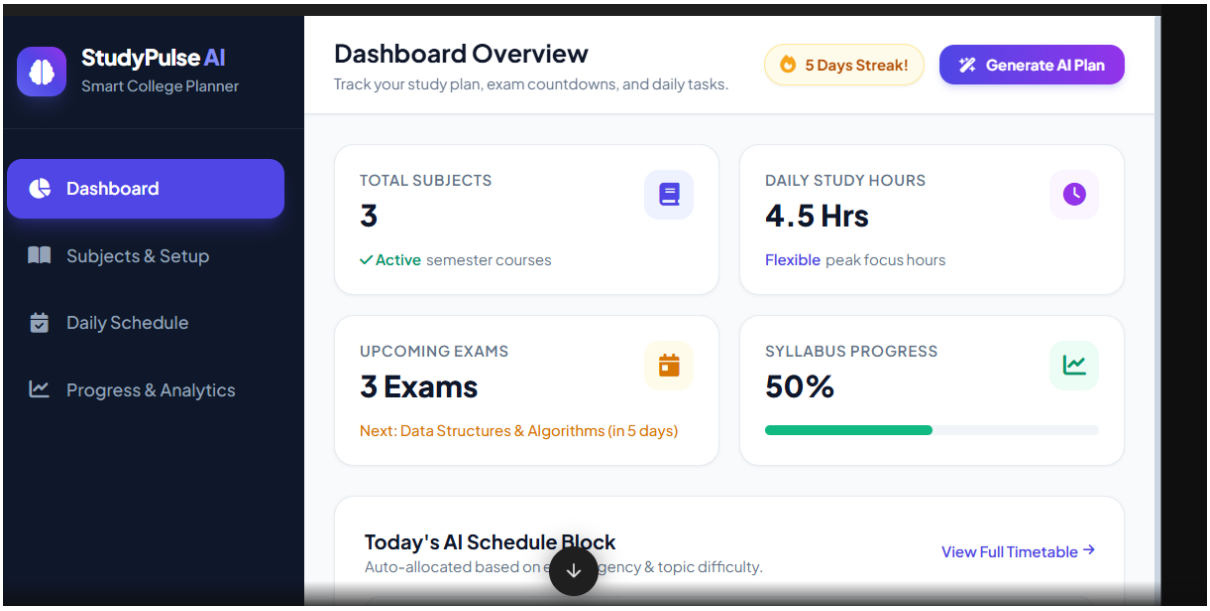
Prompt 3 :

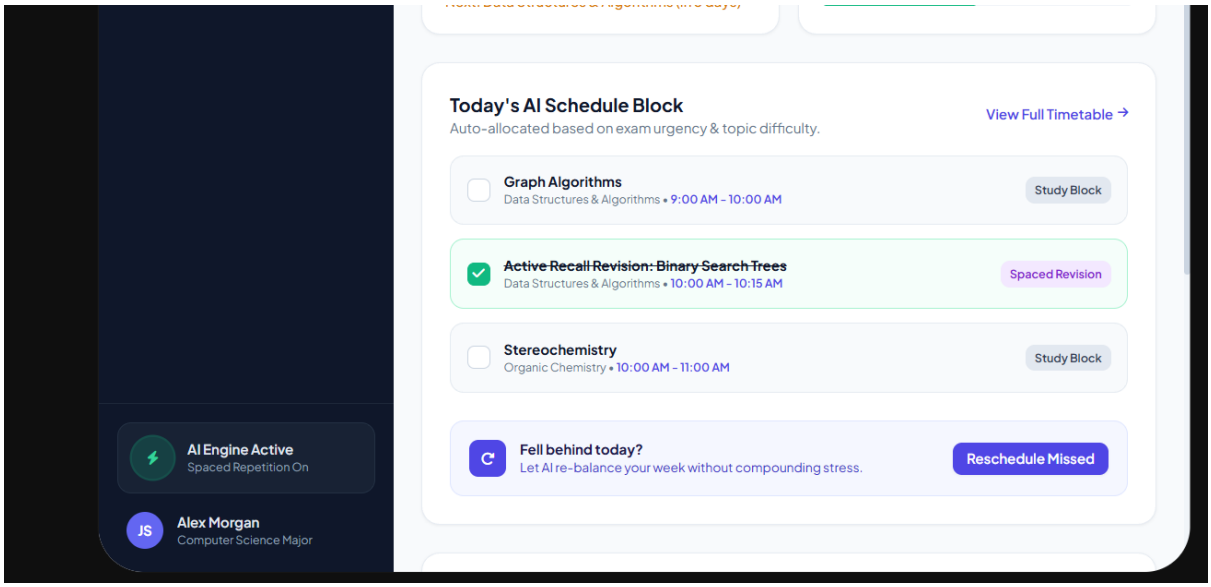
Purpose: To design a simple, attractive, and user-friendly interface.

Prompt : Design a modern and responsive AI Study Planner web application. Include a dashboard, subject input form, available study hours, exam date, priority selection, Generate Study Plan button, daily timetable, task checklist, and progress indicator. Use a clean layout, readable fonts, suitable colors, and mobile-friendly design. Make the interface easy for college students to use.



- **UI design screenshot**

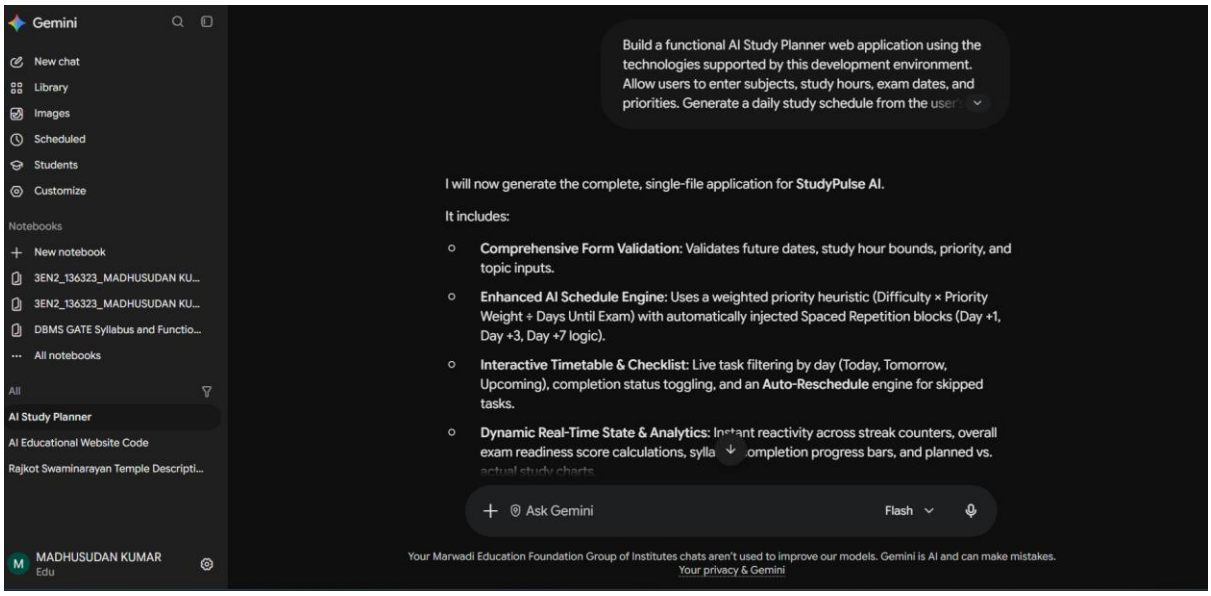


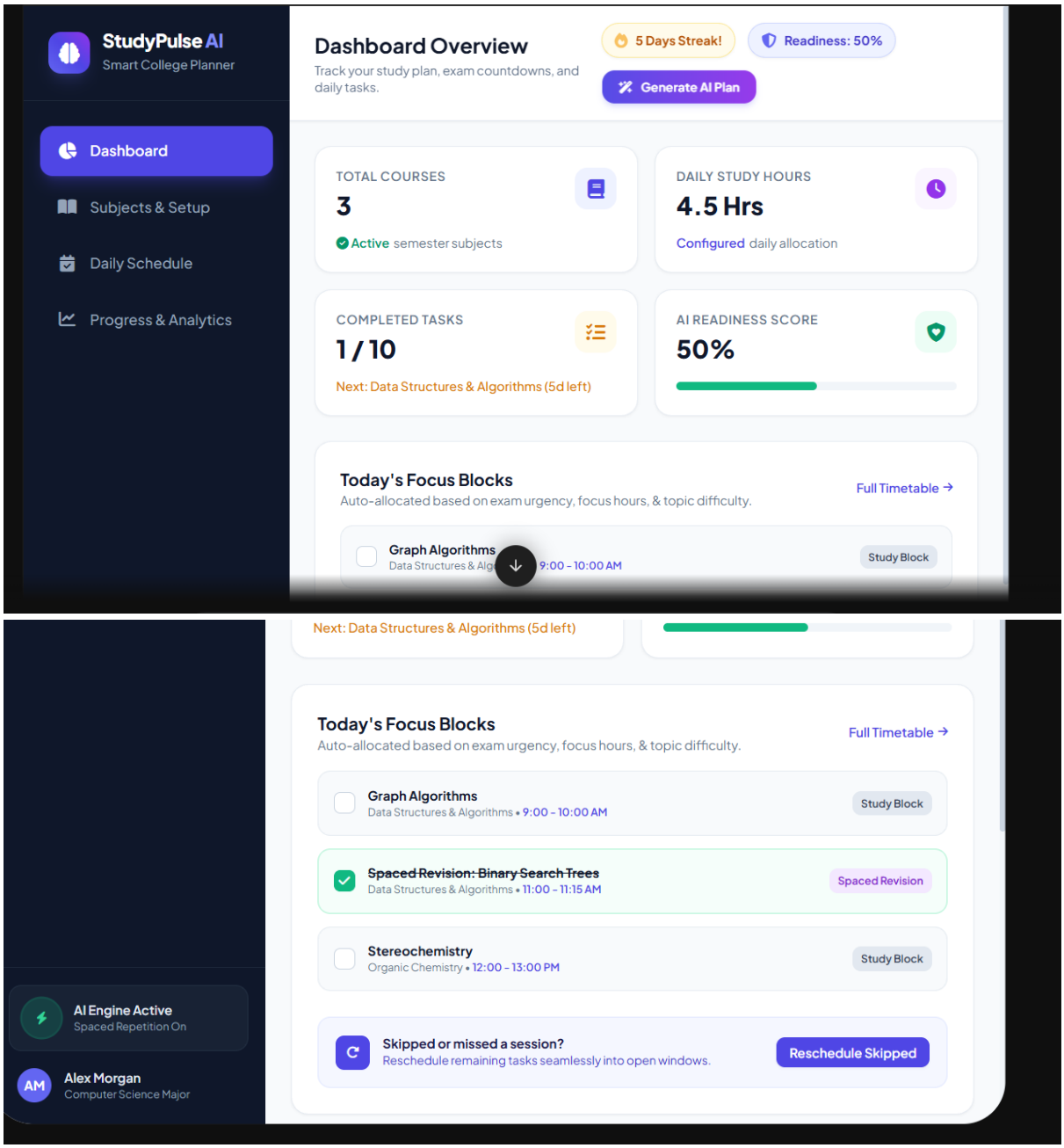


Prompt 4 :

Purpose: To create the working application.

Prompt : Build a functional AI Study Planner web application using the technologies supported by this development environment. Allow users to enter subjects, study hours, exam dates, and priorities. Generate a daily study schedule from the user's inputs, display tasks in an organized timetable, allow tasks to be marked as completed, and show study progress. Add input validation and a responsive user interface. Ensure all buttons work correctly.

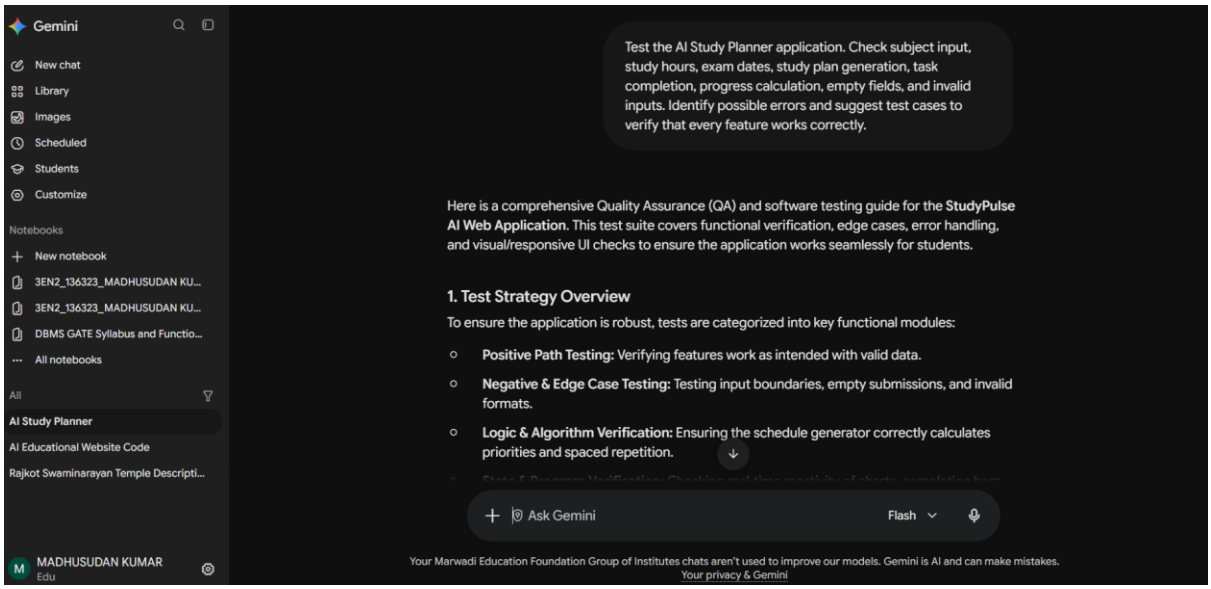


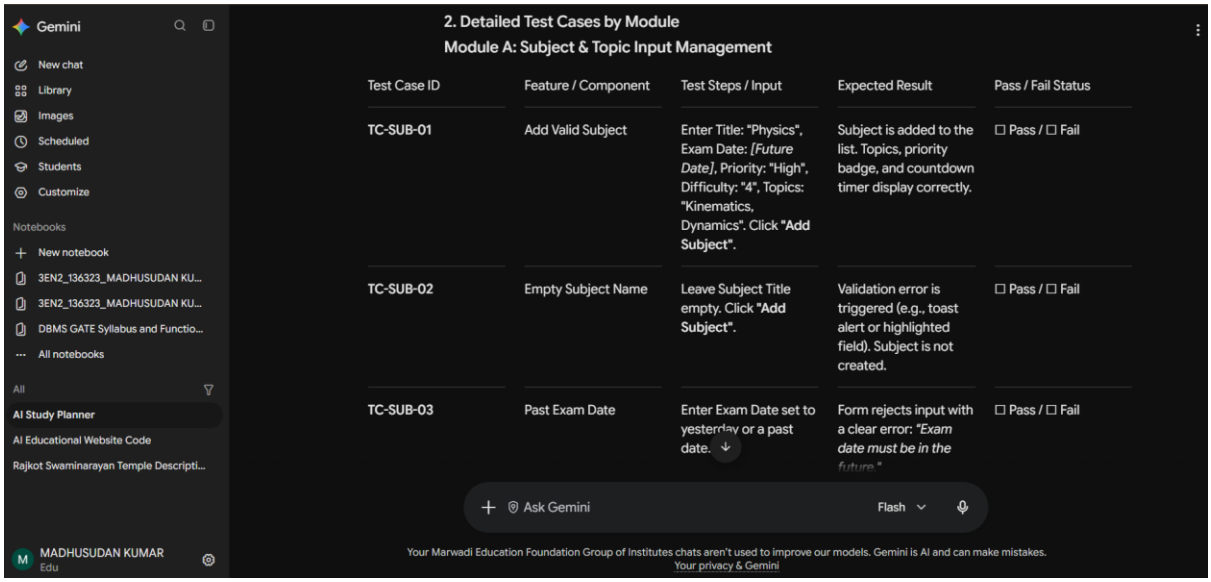


Pr ompt 5 :-

Purpose : To verify that the application works correctly.

Prompt : Test the AI Study Planner application. Check subject input, study hours, exam dates, study plan generation, task completion, progress calculation, empty fields, and invalid inputs. Identify possible errors and suggest test cases to verify that every feature works correctly.

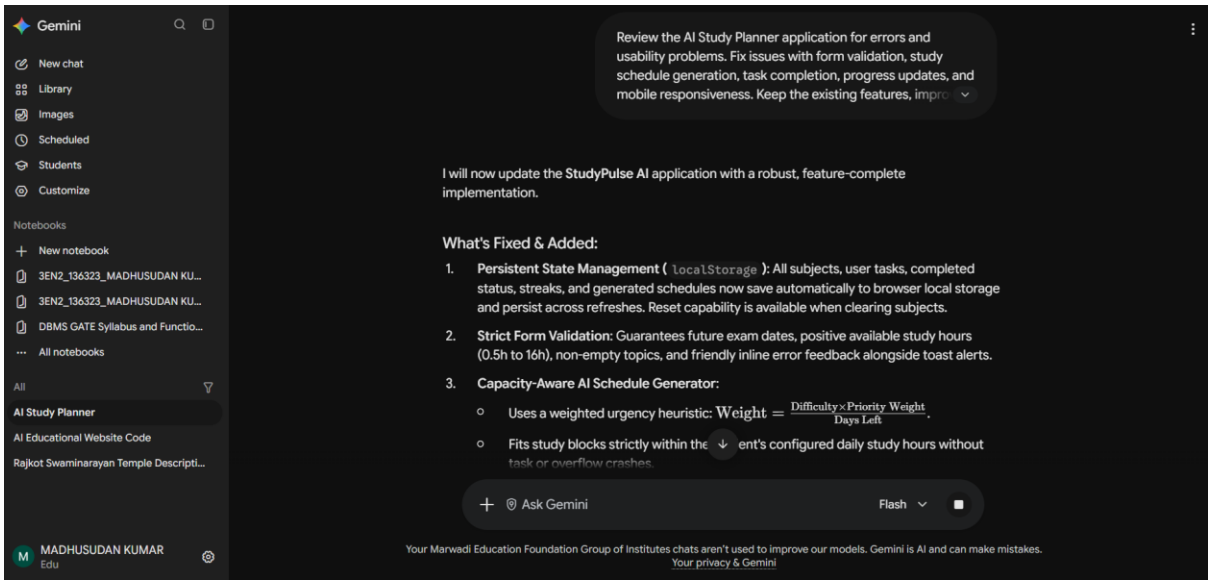




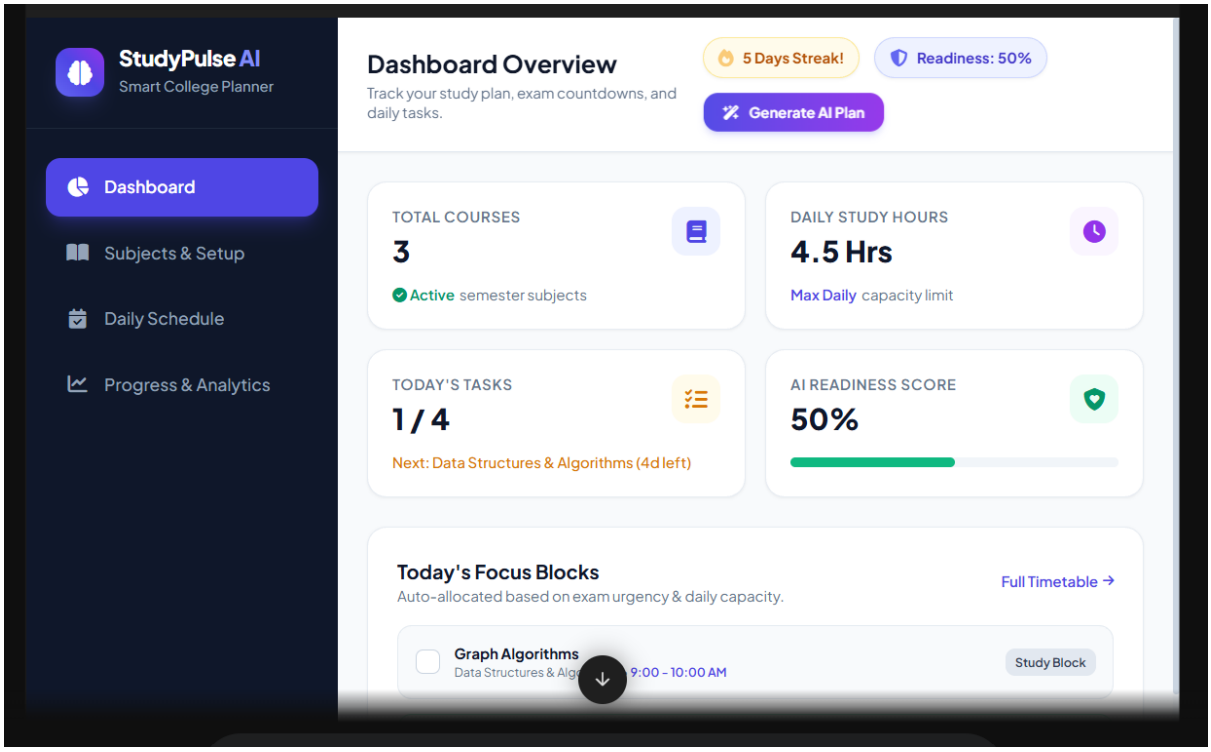
Pr ompt 6 :-

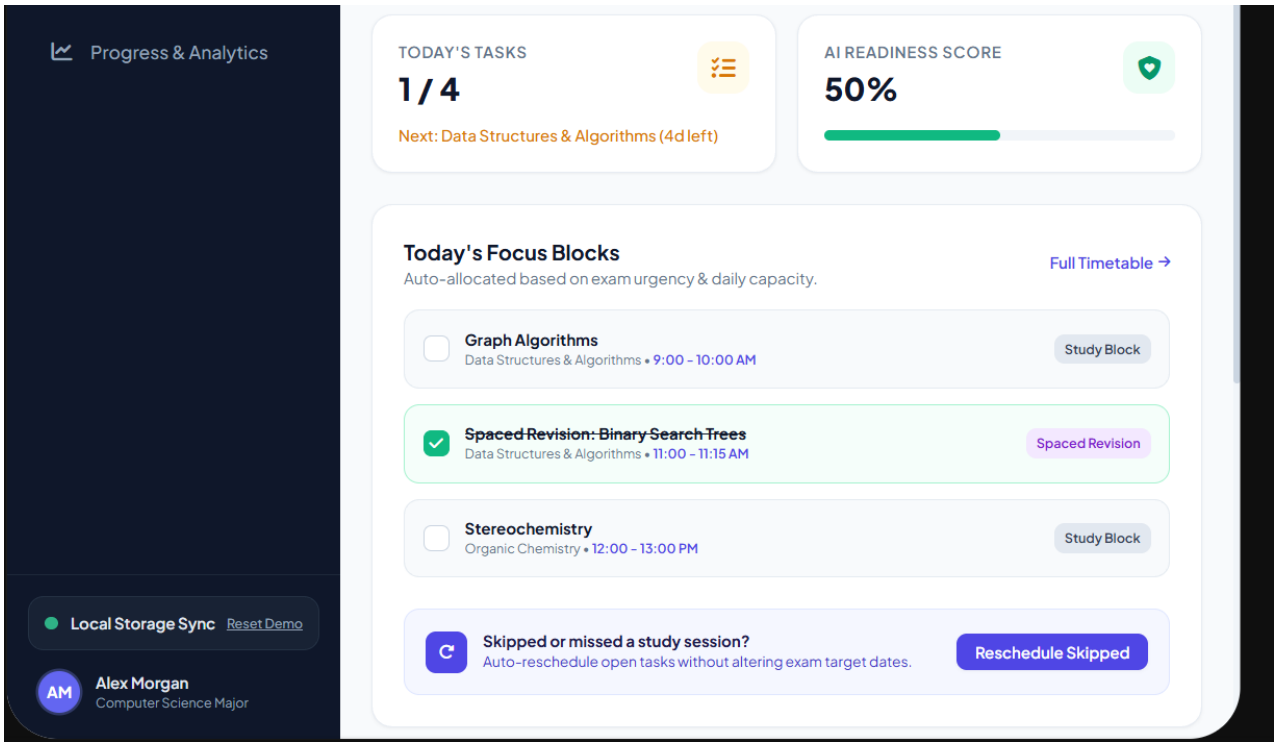
Purpose : To correct errors and improve the application.

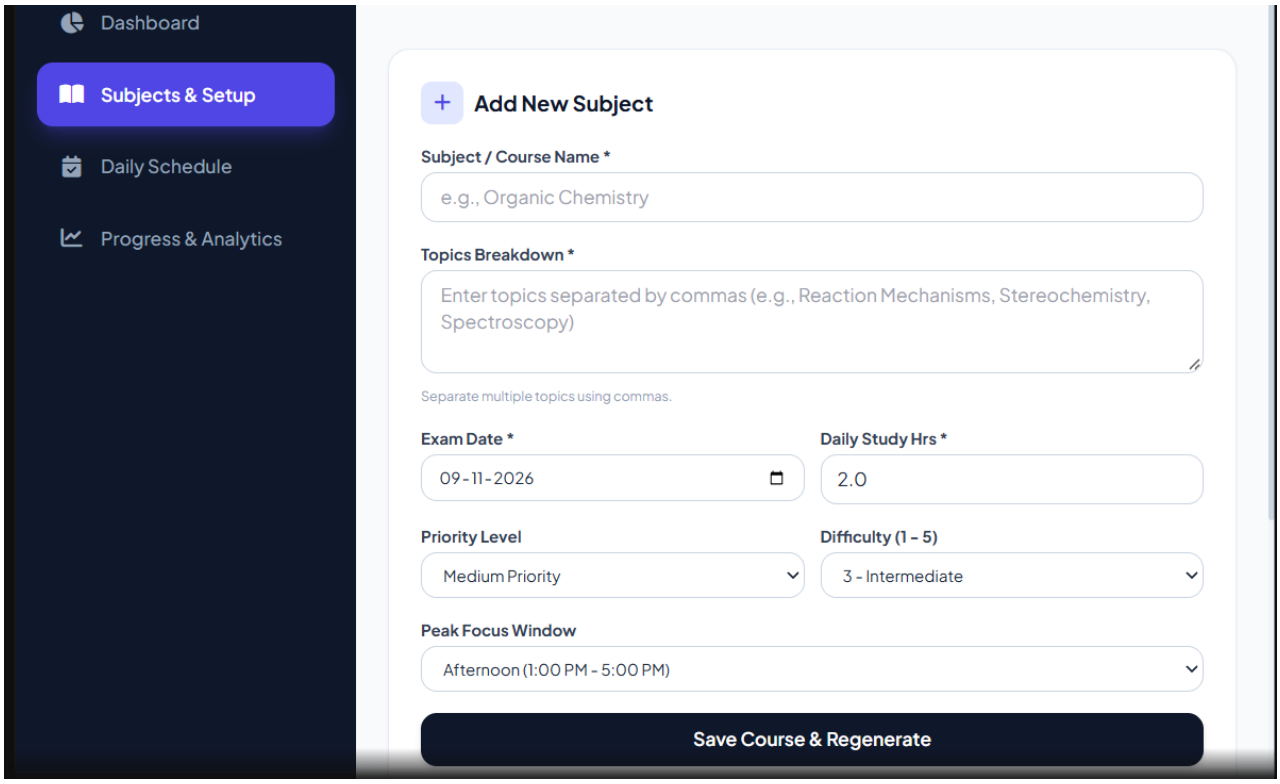
Prompt : Review the AI Study Planner application for errors and usability problems. Fix issues with form validation, study schedule generation, task completion, progress updates, and mobile responsiveness. Keep the existing features, improve the design, and ensure that the application works properly after the changes.

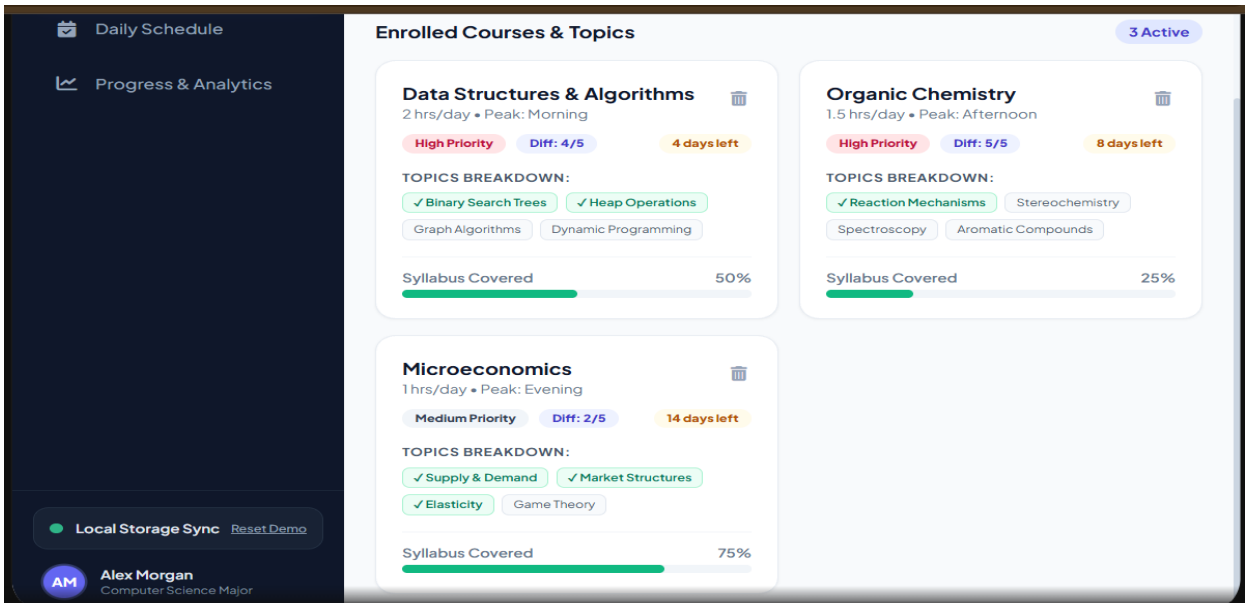


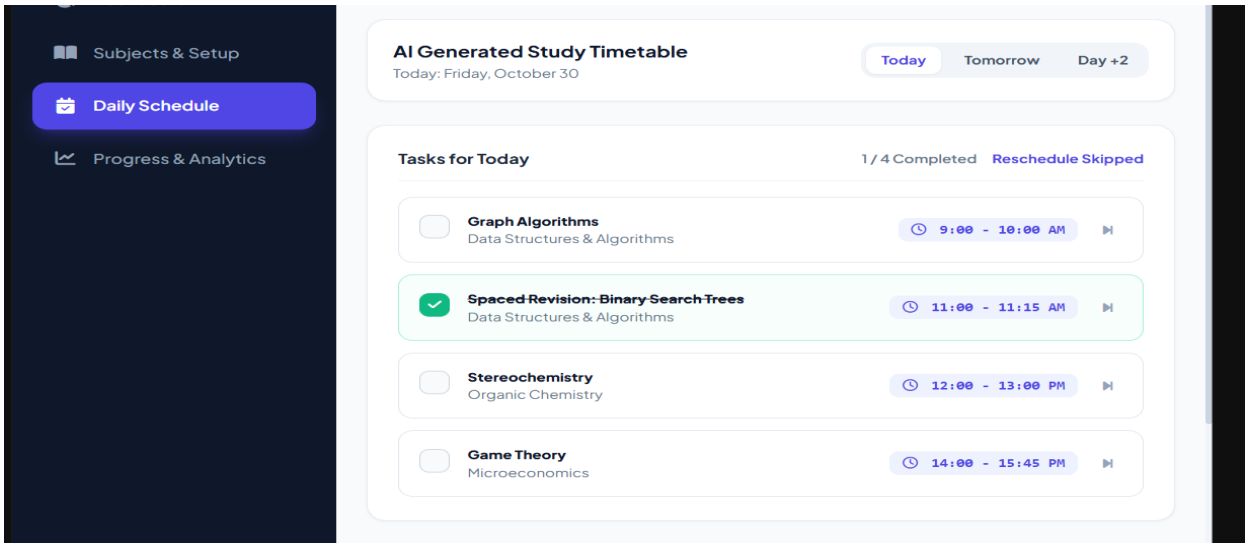
- **Home page of website**

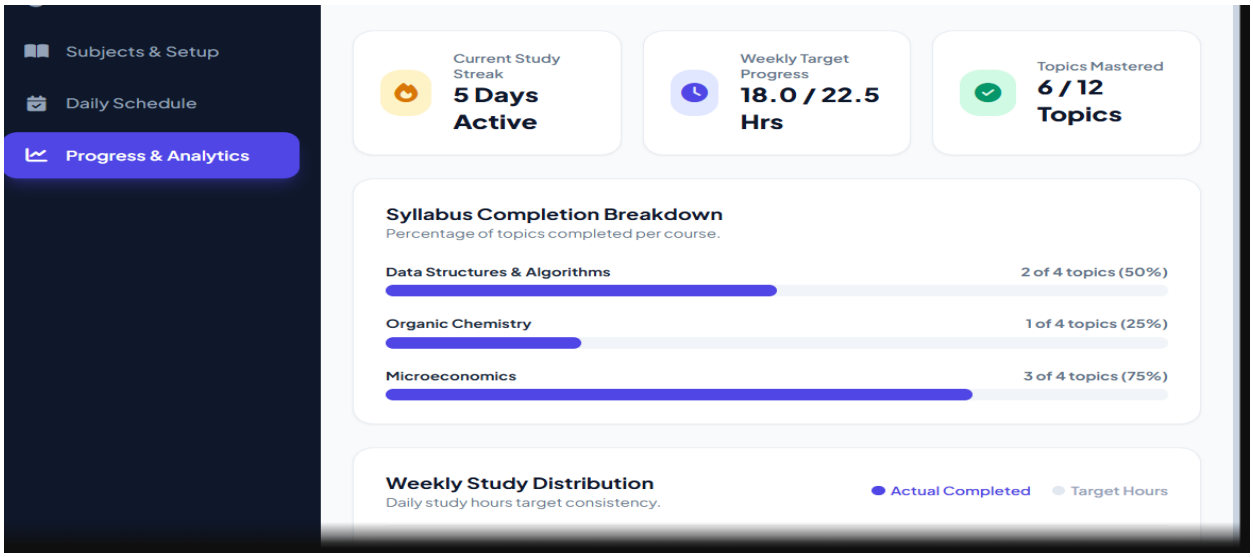


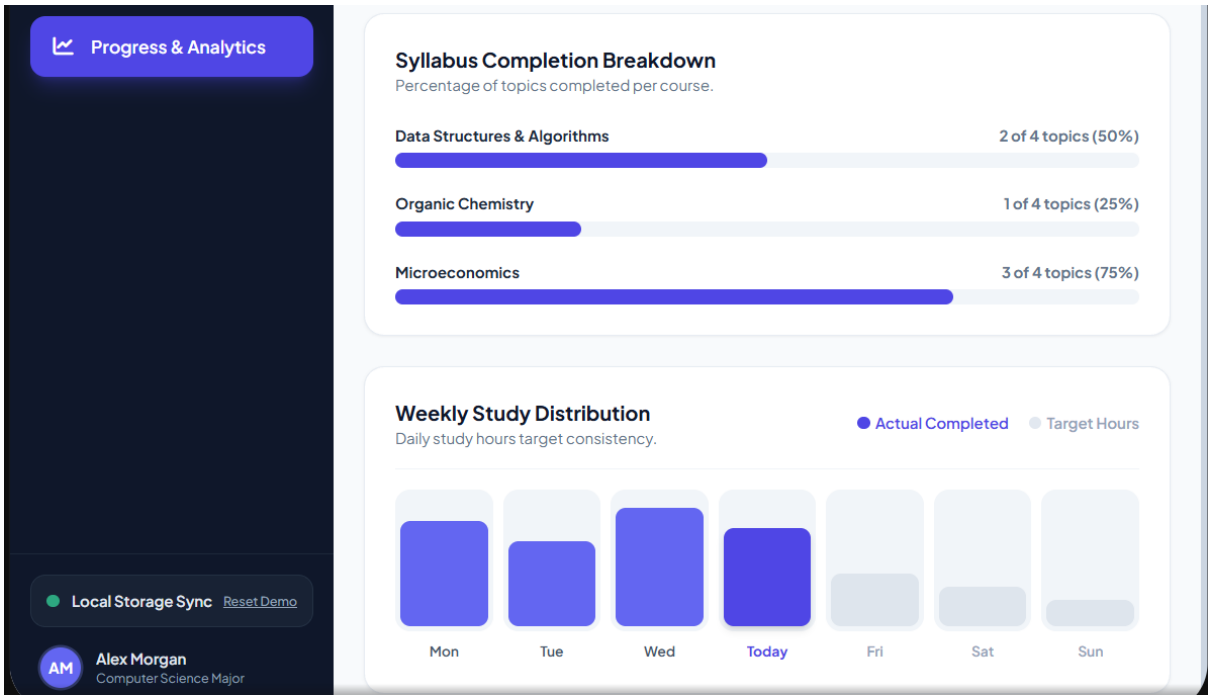

Progress & Analytics
TODAY'S TASKS
1 / 4
Next: Data Structures & Algorithms (4d left)
AI READINESS SCORE
50%
Today's Focus Blocks
Auto-allocated based on exam urgency & daily capacity.
Full Timetable →
Graph Algorithms
Data Structures & Algorithms • 9:00 - 10:00 AM
Study Block
Spaced Revision: Binary Search Trees
Data Structures & Algorithms • 11:00 - 11:15 AM
Spaced Revision
Stereochemistry
Organic Chemistry • 12:00 - 13:00 PM
Study Block
Skipped or missed a study session?
Auto-reschedule open tasks without altering exam target dates.
Reschedule Skipped
Local Storage Sync
Reset Demo
AM
Alex Morgan
Computer Science Major


Dashboard
Subjects & Setup
Daily Schedule
Progress & Analytics
Add New Subject
Subject / Course Name *
e.g., Organic Chemistry
Topics Breakdown *
Enter topics separated by commas (e.g., Reaction Mechanisms, Stereochemistry, Spectroscopy)
Separate multiple topics using commas.
Exam Date *
09-11-2026
Daily Study Hrs *
2.0
Priority Level
Medium Priority
Difficulty (1 - 5)
3 - Intermediate
Peak Focus Window
Afternoon (1:00 PM - 5:00 PM)
Save Course & Regenerate


Daily Schedule
Progress & Analytics
Enrolled Courses & Topics
3 Active
Data Structures & Algorithms
2 hrs/day • Peak: Morning
High Priority
Diff: 4/5
4 days left
TOPICS BREAKDOWN:
✓ Binary Search Trees
✓ Heap Operations
Graph Algorithms
Dynamic Programming
Syllabus Covered
50%
Organic Chemistry
1.5 hrs/day • Peak: Afternoon
High Priority
Diff: 5/5
8 days left
TOPICS BREAKDOWN:
✓ Reaction Mechanisms
Stereochemistry
Spectroscopy
Aromatic Compounds
Syllabus Covered
25%
Microeconomics
1 hrs/day • Peak: Evening
Medium Priority
Diff: 2/5
14 days left
TOPICS BREAKDOWN:
✓ Supply & Demand
✓ Market Structures
✓ Elasticity
Game Theory
Syllabus Covered
75%
Local Storage Sync
Reset Demo
AM
Alex Morgan
Computer Science Major


Subjects & Setup
Daily Schedule
Progress & Analytics
AI Generated Study Timetable
Today: Friday, October 30
Today
Tomorrow
Day +2
Tasks for Today
1 / 4 Completed
Reschedule Skipped
Graph Algorithms
Data Structures & Algorithms
9:00 - 10:00 AM
Spaced Revision: Binary Search Trees
Data Structures & Algorithms
11:00 - 11:15 AM
Stereochemistry
Organic Chemistry
12:00 - 13:00 PM
Game Theory
Microeconomics
14:00 - 15:45 PM


Subjects & Setup
Daily Schedule
Progress & Analytics
Current Study Streak
5 Days Active
Weekly Target Progress
18.0 / 22.5 Hrs
Topics Mastered
6 / 12 Topics
Syllabus Completion Breakdown
Percentage of topics completed per course.
Data Structures & Algorithms
2 of 4 topics (50%)
Organic Chemistry
1 of 4 topics (25%)
Microeconomics
3 of 4 topics (75%)
Weekly Study Distribution
Daily study hours target consistency.
Actual Completed
Target Hours



studypulse-ai-web-application
https://sites.google.com/marwadiuniversity.ac.in/studypulse-ai-web-application/full-page-embed?authuser=1

7. Conclusion

The AI Study Planner project demonstrates how artificial intelligence tools and prompt engineering can be used to develop a practical application for students. The project focuses on solving common academic problems such as poor time management, difficulty organizing subjects, and inconsistent study routines.

AI tools can assist with idea generation, requirement analysis, user interface design, code generation, testing, and debugging. By using clear prompts and improving the results through repeated testing, the development process can become more organized and efficient.

The proposed application provides a structured way to prepare study schedules and track task completion. In the future, the application could be improved by adding reminders, calendar integration, automatic timetable adjustment, performance analytics, and integration with a reliable AI service for more personalized recommendations.

In conclusion, the AI Study Planner is a useful educational project that demonstrates the practical application of AI-assisted software development and prompt engineering.